Intrusion Detection System

What is IDS

- It is system that combine the use of both software and hardware to monitor, identify and assess possible malicious actions that compromise the CIA traid.
- These malicious actions can take place in network, host machine, application.
- The responsibility of the IDS is to be able to accurately detect and notify system manager that it found an abnormal event.
- It is passive scanning, and set to default to pass

Method of detection

- Misuse detection
  - It is a signature and knowledge based detection method. By accessing an database that stored all well-known attack signature and matching traffic pattern with these signature to detect intrusion
- Anomaly detection
  - It identify the deviation from normal usage pattern and notify an alert when the activity, such as CPU usage, volume of network traffic, is outside the baseline parameters.

How do we evalute and measure the performance

We use:

- Precision: (TP)/(TP+FP)
- Recall: (TP)/(TP+FN)
- F1-score: it is the harmonic mean between precision and recall, providing a balanced measure to describe the overall performance of the IDS

Type of IDS

Base on the environment of the IDS being deployed, we distinguish them into 3 categories

- Host based
- Network based
- Application based

Host based IDS

- It runs on a particular computer and machine, monitoring activity only on that system
- Advantage:
  - Detect local event
  - Able to view encrypted data as the machine will decrypt it
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- Disadvantage:
  - It is vulnerable to attack against specific OS
  - It is hard to manage, as it is base on configuration management. You need to have completely understand what the system is doing before configure it.

Network base IDS

- It reside on a network segment that allow it to monitor the traffic of a specific network segment
- It filter out any malicious or suspicious network packet before it reaches to the nodes.
- Advantage
  - It can monitor large network with limited device
  - Minimal disrupts operations
- Disadvantage
  - It can be overwhelm by the volume of traffic and unable to filter out network packet
  - It is unable to analyse encrypted packet

Application based IDS

- It focus on looking for abnormal activity at application
- It will look at the file system, network and memory to ensure the usage of computer resource is normal and no authorized program is running.
- Advantage:
    - It can observe the interaction between an application and users
    - Able to work with encrypted data
- Disadvantage:
    - It is more susceptible to tampering and spoofing attack, as it has limited context for comparison
    - It does not provide a full picture of the system